LAB 03

Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Ingestion
TSC reference	A6, K9
Duration	20 min
Folder	labs_fullcode/Lab03-alpaca-market-data/
Files	Lab03-alpaca-market-data/ingest_stage.py, pipeline/ingest.py, pipeline/reference/sample_headlines.json, Lab03-alpaca-market-data/solution/ingest.py
Prompts	Lab03_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

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Scenario

The pipeline's research and trading stages must all read the same clean data. The desk's data engineer automates ingestion: bars for the whole universe at four timeframes, a data-quality report that blocks bad files, a cache every later stage reads, and a first unstructured source - news headlines - that will later feed a sentiment agent.

Objective

Build `python -m pipeline ingest` to fetch and cache daily, weekly, monthly and hourly bars for the universe, run automated data-quality checks, ingest headlines as unstructured data, and describe the forms of data and the 3 Vs of big data.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 03	python3 Lab03-alpaca-market-data/ingest_stage.py (same as: python -m pipeline ingest)
The complete code	pipeline/ingest.py (reference copy: Lab03-alpaca-market-data/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab03-alpaca-market-data/Lab03_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Understand how bars are requested (K9)

Open common/data.py and read get_bars(): online it builds a StockBarsRequest for StockHistoricalDataClient; offline it reads data/<SYMBOL>.csv and resamples weekly and monthly bars.

AI prompt - paste into your AI coding assistant
Read labs_fullcode/common/data.py and explain in five bullet points how get_bars() requests bars from Alpaca (TimeFrame, adjustment, feed, time zone) and what it does offline. Do not change the file.

Check: You can state that online requests use adjustment=ALL and feed=IEX, and that offline data are the bundled Yahoo Finance CSVs.

Step 2: Build the cache and load_bars() (A6, K9)

Ask your assistant for the cache helpers at the top of pipeline/ingest.py.

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/ingest.py with cache_dir() (config data.cache_dir under labs), cache_file(symbol, timeframe) -> _cache/bars/<SYMBOL>_<TIMEFRAME>.csv, and load_bars(symbol, start=None, end=None, timeframe='1Day') that reads the cached CSV (hourly timestamps as America/New_York, daily as dates), slices start and end inclusively (weekly and monthly bars are labelled with the period end, so allow 6 / 30 days after end), and falls back to common.data.get_bars when the file is missing or when online with end=None. Add load_closes(symbols, start, end) that inner-joins daily closes like common.data.get_closes. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python3 -c "from pipeline.ingest import load_bars; print(load_bars('SPY','2025-12-01','2025-12-31').tail(1))" prints the 2025-12-31 bar with close 676.64 (from the data source until the cache exists).

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Step 3: Add automated data-quality checks (A6)

Ask your assistant for quality_checks() and the blocking rules.

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/ingest.py add quality_checks(df, timeframe, end, max_move) returning a dict: rows, first, last, missing_values, duplicate_timestamps, not_sorted, high_below_low, close_outside_range, zero_volume, gaps_over_4_days (daily only), big_moves (daily |return| > max_move, from config data.max_abs_daily_move) and stale (last daily bar more than 5 days before end). BLOCKING = missing_values, duplicate_timestamps, not_sorted, high_below_low, close_outside_range, stale. Status is FAIL if any blocking count is non-zero (the file is NOT cached), CHECK if there are big moves or zero-volume bars, else OK. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: quality_checks() returns 0 for every blocking check on the bundled SPY daily file.

Step 4: Ingest an unstructured source: news headlines (A6, K9)

Ask your assistant for the headline functions. Offline they use the SYNTHETIC sample in pipeline/reference/sample_headlines.json (training text, not real news).

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/ingest.py add _fetch_headlines(symbols): online use alpaca.data.historical.news.NewsClient with NewsRequest(symbols=..., start=now - 7 days, limit=10) and keys from common.config.get_keys(); on any error, or offline, load pipeline/reference/sample_headlines.json and set _source to 'bundled SYNTHETIC sample (training only - not real news)'. Add load_headlines(symbols=None) that reads _cache/news/headlines.json when it exists. Treat headline text as untrusted data. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: load_headlines(['NVDA']) returns 4 headline dictionaries with keys date, symbol and headline.

Step 5: Build the ingest run with the report (A6, K9)

Ask your assistant for run() and main().

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/ingest.py add run(symbols=None, timeframes=None) for `python -m pipeline ingest`: symbols = universe trade + pair legs + benchmark; for each symbol and config timeframe fetch bars with common.data.get_bars (1Hour uses data.hourly_window; record 'not available offline' on FileNotFoundError), run quality_checks, cache every non-FAIL file, write _outputs/Lab03_data_quality.csv and _cache/manifest.json (ingested_at, source, rows/first/last/status per file, headline count), save the headlines to _cache/news/headlines.json, print the daily quality table, bars per timeframe, headline count by symbol, SPY and AAPL annualised mean, volatility, skew, excess kurtosis and correlation, a DATA_FORMS table (bars, quotes, trades, corporate actions, fundamentals = structured; news, alternative data = unstructured) and three lines on Volume, Velocity and Variety. Save chart Lab03_market_data.png (universe growth of 100 on a log scale, SPY/AAPL return histograms, SPY 21-day volatility, SPY hourly closes). Exit code 1 if any file FAILs. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline ingest --help shows --symbols and --timeframes.

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Step 6: Run the ingestion stage (A6, K9)

Run: python -m pipeline ingest (same as: python3 Lab03-alpaca-market-data/ingest_stage.py).

Check: 12 symbols x 4 timeframes; daily files 2,011 rows each (2018-01-02 to 2025-12-31); 1Day 24,132 bars, 1Week 5,016 (418 per symbol), 1Month 1,152 (96), 1Hour 440 (SPY only offline - 11 symbols 'not available offline'); status CHECK for big moves (NVDA 40, AAPL 12, JPM 12, BAC 11, SPY 5) and OK for GLD and TLT; 12 SYNTHETIC headlines; SPY mean 15.14%, vol 19.46%, excess kurtosis 13.12; AAPL 28.66% / 30.79%; correlation 0.77; 'RESULT: 37 files cached, 0 failed quality checks'.

Step 7: Inspect the cache and the manifest (A6)

Open _cache/manifest.json and _outputs/Lab03_data_quality.csv. Rerun the Step 2 check command - it now reads from the cache. Look at _outputs/Lab03_market_data.png.

Check: manifest.json lists 37 files with rows, first, last and status; the SPY 2025-12-31 close is still 676.64 (the cache is identical to the source).

Step 8: Design a big-data extension (A6)

Decide which additional data form would most improve the desk's pipeline and what governance it needs.

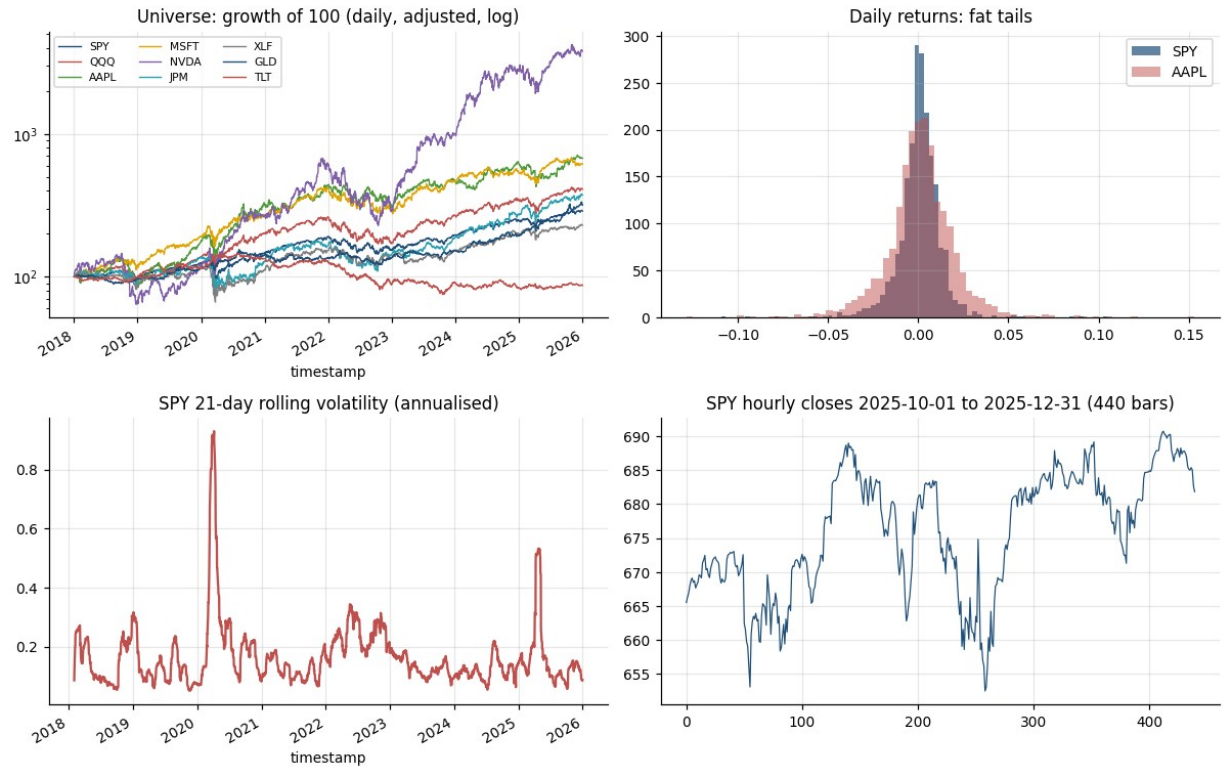
AI prompt - paste into your AI coding assistant
Using the DATA_FORMS table in labs_fullcode/pipeline/ingest.py, propose how to add Alpaca news for the full trade universe and 1-minute bars for 500 US stocks as scheduled ingest jobs: storage format, refresh frequency, deduplication key, estimated rows per year, and three governance controls (data licence, point-in-time universe, prompt-injection risk when headlines reach an AI agent). Answer in a short table; do not write code yet.

Check: Your table estimates about 49 million one-minute rows a year for 500 stocks and names at least three controls.

Expected Output (Full Code)

python -m pipeline ingest (offline): 37 files cached in _cache/bars, 0 failed quality checks; daily 2,011 bars per symbol, weekly 418, monthly 96, SPY hourly 440 (Q4 2025); big-move flags NVDA 40, AAPL 12, SPY 5; 12 synthetic headlines cached; SPY annualised mean 15.14% and volatility 19.46% (excess kurtosis 13.12), AAPL 28.66% / 30.79%, correlation 0.77; files _outputs/Lab03_data_quality.csv, _cache/manifest.json and chart _outputs/Lab03_market_data.png.

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Lab 3 output (labs_fullcode/_outputs/Lab03_market_data.png), offline mode on the bundled data.

Debrief

Ingestion is the first automated stage: it fixes one version of the truth for every later stage and blocks bad data before a model can learn from it. Bars, quotes, trades, news and alternative data are different forms of financial data (K9) with different volume, velocity and variety - which is what turns a trading data set into a big-data application (A6).

Troubleshooting

Symptom	What to do
Hourly bars 'not available offline' for most symbols	Expected: only SPY hourly data are bundled; online Alpaca returns hourly bars for every symbol.
Alpaca returns fewer hourly bars than expected	The free IEX feed covers only IEX trades; use daily bars or a SIP subscription for full volume.
Status CHECK on many symbols	CHECK means 'look at the big moves' (earnings, splits, 2020 crash); only FAIL blocks a file.
Later stages ignore new data	load_bars reads the cache offline; rerun python -m pipeline ingest after adding data.

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